7.7 Pre-Quiz: Logarithms — Markscheme

1. [Maximum mark: 5]

   (a) $\ln 3 + \ln 4 = \ln(3 \cdot 4) = \ln 12.$ **A1**

   (b) $3 \ln 2 = \ln 2^3$ **A1**

   $= \ln 8.$ **A1**

   (c) $-\ln \dfrac{1}{2} = \ln\left(\dfrac{1}{2}\right)^{-1}$ **A1**

   $= \ln 2.$ **A1**

   Accept also $-\ln \frac{1}{2} = -(\ln 1 - \ln 2) = -(-\ln 2) = \ln 2.$

2. [Maximum mark: 5]

$\log_{10} 2 = p$ and $\log_{10} 3 = q$.

(a) $\log_{10} 24 = \log_{10}(8 \cdot 3) = \log_{10} 2^3 + \log_{10} 3$ **M1**

$= 3\log_{10} 2 + \log_{10} 3$ **A1**

$= 3p + q.$ **A1**

(b) Change of base: $\log_3 8 = \dfrac{\log_{10} 8}{\log_{10} 3} = \dfrac{3\log_{10} 2}{\log_{10} 3}.$ **A1**

$= \dfrac{3p}{q}.$ **A1**

3. [Maximum mark: 5]

$\log_{10} a = \dfrac{1}{3}$, where $a > 0$.

(a) $\log_{10}\left(\dfrac{1}{a}\right) = \log_{10} a^{-1} = -\log_{10} a$ **A1**

$= -\dfrac{1}{3}$. **A1**

(b) Change of base: $\log_{1000} a = \dfrac{\log_{10} a}{\log_{10} 1000} = \dfrac{\log_{10} a}{3}$. **A1**

$= \dfrac{1/3}{3}$ **A1**

$= \dfrac{1}{9}$. **A1**

4. [Maximum mark: 5]

$2\ln x = \ln 9 + 4$, with $x = p\,e^q$, $p, q \in \mathbb{Z}^+$.

$2\ln x - \ln 9 = 4$. Use $m\ln x = \ln x^m$: $\ln x^2 - \ln 9 = 4$. **M1**

Use $\ln a - \ln b = \ln \dfrac{a}{b}$: $\ln \dfrac{x^2}{9} = 4$. **M1**

Exponentiate: $\dfrac{x^2}{9} = e^4 \;\Rightarrow\; x^2 = 9e^4$. **A1**

Since $x > 0$: $x = \sqrt{9e^4} = 3e^2$. **A1**

So $p = 3,\; q = 2$. **A1**

Accept any valid method, including expressing $4 = 4\ln e$ *first to combine all terms as a single log before exponentiating.*

5. [Maximum mark: 5]

$3\log_8(10x) - \log_4 x = 1,\ \ x > 0.$

Change of base to base 2: $\log_8 y = \dfrac{\log_2 y}{3}, \quad \log_4 y = \dfrac{\log_2 y}{2}.$ **M1**

Substitute: $3 \cdot \dfrac{\log_2(10x)}{3} - \dfrac{\log_2 x}{2} = 1,$ so $\log_2(10x) - \dfrac{1}{2}\log_2 x = 1.$ **A1**

Apply power rule and quotient rule: $\log_2(10x) - \log_2 x^{1/2} = \log_2\left(\dfrac{10x}{\sqrt{x}}\right) = \log_2(10\sqrt{x}) = 1.$ **M1**

Power rule and quotient rule may be applied in any order; this M1 rewards either rewriting toward a single logarithm with the same base.

Equate arguments: $10\sqrt{x} = 2,$ so $\sqrt{x} = \dfrac{1}{5}.$ **A1**

$x = \dfrac{1}{25}.$ **A1**

6. [Maximum mark: 14]

$f(x) = \log_2(8x),\;\; x > 0.$

(a)(i) $f(2) = \log_2(8 \cdot 2) = \log_2 16.$ **M1**

$= 4.$ **A1**

(a)(ii) $f\left(\dfrac{1}{8}\right) = \log_2\left(8 \cdot \frac{1}{8}\right) = \log_2 1 = 0.$ **A1**

(b) Set $y = \log_2(8x)$ and swap x and y: $x = \log_2(8y)$. **M1**

Rewrite as exponential: $2^x = 8y$. **M1**

$y = \dfrac{2^x}{8}$ **A1**

$f^{-1}(x) = \dfrac{2^x}{8} \quad (= 2^{x-3}).$ **A1**

Accept either form; $\dfrac{2^x}{8}$ *and* 2^{x-3} *are equivalent.*

(c) $f^{-1}(0) = \dfrac{2^0}{8} = \dfrac{1}{8}.$ **A1**

(d) $y = f(4x^2) = \log_2(8 \cdot 4x^2) = \log_2(32x^2).$

Use addition rule for logs: $\log_2(32x^2) = \log_2 32 + \log_2 x^2.$ **M1**

$= 5 + \log_2 x^2.$ **A1**

Use exponent property: $\log_2 x^2 = 2\log_2 x.$ **M1**

So $f(4x^2) = 5 + 2\log_2 x.$ **A1**

Compared with $y = \log_2 x$: vertical stretch (dilation) by scale factor 2, then vertical translation 5 units upward. **A2**

Award A2 for both transformations stated with the correct order. Award A1 if the two transformations are correct but the order is not specified or is reversed.